

21-2b Four Properties of Indifference Curves

Because indifference curves represent a consumer's preferences, they have certain properties that reflect those preferences. Here we consider four properties that describe most indifference curves:

- *Property 1: Higher indifference curves are preferred to lower ones.* People usually prefer to consume more rather than less. This preference for greater quantities is reflected in the indifference curves. As [Figure 3](#) shows, higher indifference curves represent larger quantities of goods than lower indifference curves. Thus, a consumer prefers being on higher indifference curves.
- *Property 2: Indifference curves slope downward.* The slope of an indifference curve reflects the rate at which a consumer is willing to substitute one good for the other. In most cases, the consumer likes both goods. Therefore, if the quantity of one good decreases, the quantity of the other good must increase for the consumer to be equally happy. For this reason, most indifference curves slope downward.
- *Property 3: Indifference curves do not cross.* To see why this is true, suppose that two indifference curves did cross, as in [Figure 4](#). Then, because point A is on the same indifference curve as point B, the two points would make the consumer equally happy. In addition, because point B is on the same indifference curve as point C, these two points would make the consumer equally happy. But these conclusions imply that points A and C would also make the consumer equally happy, even though point C has more of both goods. This contradicts our assumption that the consumer always prefers more of both goods to less. Thus, indifference curves cannot cross.

Figure 4

The Impossibility of Intersecting Indifference Curves

A situation like this can never happen. According to these indifference curves, the consumer would be equally satisfied at points A, B, and C, even though point C has more of both goods than point A.

- *Property 4: Indifference curves are bowed inward.* The slope of an indifference curve is the marginal rate of substitution—the rate at which the consumer is willing to trade off one good for the other. The marginal rate of substitution (*MRS*) usually depends on the amount of each good the consumer is currently consuming. In particular, because people are more willing to trade away goods that they have in abundance and less willing to trade away goods of which they have little, the indifference curves are bowed inward toward the graph's origin. As an example, consider [Figure 5](#). At point A, the consumer has a lot of Pepsi and only a little pizza, so she is very hungry but not very thirsty. To willingly give up 1 pizza, she would have to receive 6 liters of Pepsi: The *MRS* is 6 liters of Pepsi per pizza. By contrast, at point B, the consumer has little Pepsi and a lot of pizza, so she is very thirsty but not very hungry. At this point, she would be willing to give up 1 pizza to get 1 liter of Pepsi: The *MRS* is 1 liter of Pepsi per pizza. Thus, the bowed shape of the indifference curve reflects the consumer's greater willingness to give up a good that she already has in abundance.

Figure 5

Bowed Indifference Curves

Indifference curves are usually bowed inward. This shape implies that the marginal rate of substitution (*MRS*) depends on the quantity of the two goods the consumer is currently consuming. At point A, the consumer has little pizza and much Pepsi, so she requires a lot of extra Pepsi to induce her to give up one of the pizzas: The *MRS* is 6 liters of Pepsi per pizza. At point B, the consumer has much pizza and little Pepsi, so she requires only a little extra Pepsi to induce her to give up one of the pizzas: The *MRS* is 1 liter of Pepsi per pizza.

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